

```
#include <stdio.h>
```

//Linear search function

```
int linearSearch(int m,int n,int a[m][n],int k){

    for(int i=0;i<m;i++){

        for(int j=0;j<n;j++){

            if(a[i][j]==k){

                printf("Target found at index [%d][%d]",i,j);

                return;

            }

        }

    }

}
```

//Minimum and maximum function

```
int minmax(int m,int n,int a[m][n]){

    int max=a[0][0];

    int min=a[0][0];

    for(int i=0;i<m;i++){

        for(int j=0;j<n;j++){

            if(a[i][j]>max){

                max=a[i][j];

            }

            if(a[i][j]<min){

                min=a[i][j];}

        }

    }

}
```

```
printf("\nmax=%d,min=%d",max,min);

return;

}
```

//Sum of all elements in 2D array

```
int sum(int m,int n,int a[m][n]){

    int sum=0;

    for(int i=0;i<m;i++){

        for(int j=0;j<n;j++){

            sum=sum+a[i][j];

        }

    }

    printf("\nsum of all elements =%d\n",sum);

    return;

}
```

//Upper triangle function

```
int upperTriangle(int m,int n,int a[m][n]){

    int b[m][n];

    for(int i=0;i<m;i++){

        for(int j=0;j<n;j++){

            if(i<=j){

                b[i][j]=a[i][j];

            }

            else{

                b[i][j]=0;

            }

        }

    }

}
```

```

    }

}

printf("Upper triangle:\n");

for(int i=0;i<m;i++){

    for(int j=0;j<n;j++){

        printf("%d\t",b[i][j]);

    }

    printf("\n");

}

return;

}

```

//Sum of each row function

```

int sumrow(int m,int n,int a[m][n]){

    for(int i=0;i<m;i++){

        int sumr=0;

        for(int j=0;j<n;j++){

            sumr=sumr+a[i][j];

        }

        printf("the sum of row %d = %d\n",i,sumr);

    }

    return;

}

```

//Lower triangle function

```

int lowerTriangle(int m,int n,int a[m][n]){

```

```

int b[m][n];

for(int i=0;i<m;i++){

    for(int j=0;j<n;j++){

        if(i>=j){

            b[i][j]=a[i][j];

        }

        else{

            b[i][j]=0;

        }

    }

}

printf("Lower triangle:\n");

for(int i=0;i<m;i++){

    for(int j=0;j<n;j++){

        printf("%d\t",b[i][j]);

    }

    printf("\n");

}

return;

}

```

//Sum of each column function

```

int sumcolumn(int m,int n,int a[m][n]){

    for(int i=0;i<m;i++){

        int sumc=0;

        for(int j=0;j<n;j++){

            sumc=sumc+a[j][i];

        }

    }

}

```

```

    }

    printf("the sum of column %d = %d\n",i,sumc);

}

return;

}

```

//Matrix multiplication

```

int matmult(int m,int n,int p,int q,int a[m][n],int b[p][q] ){

    if(n!=p){

        printf("matrix multiplication is not possible");

        return;

    }

    else {

        int c[m][q];

        for(int i=0;i<m;i++){

            for(int j=0;j<q;j++){

                c[i][j]=0;

                for(int k=0;k<p;k++){

                    c[i][j]+= a[i][k]*b[k][j];

                }

            }

        }

        for(int i=0;i<m;i++){

            for(int j=0;j<q;j++){

                printf("%d\t",c[i][j]);

            }

            printf("\n");

```

```
}  
  
return;  
  
}  
  
}
```

//Main function

```
int main() {  
  
    int d=3;  
  
    int e=3;  
  
    int a[3][3]={  
  
        {1,2,3},  
  
        {4,5,6},  
  
        {7,8,9} };  
  
    int target=5;  
  
    int b[3][3]={  
  
        {4,5,2},  
  
        {1,7,4},  
  
        {2,7,4}  
  
        };  
  
    int m=3,n=3,p=3,q=3;  
  
    linearSearch(d,e,a,target);  
  
    minmax(d,e,a);  
  
    sum(d,e,a);  
  
    upperTriangle(d,e,a);  
  
    printf("\n");  
  
    sumrow(d,e,a);  
  
    printf("\n");  
  
    lowerTriangle(d,e,a);
```

}

HackerRank

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DSA Lab 3 Session 2

Diagonal Difference

Exit Full Screen View

Problem

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4 5 6
9 8 9

The left-to-right diagonal = $1 + 5 + 9 = 15$.

The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the `diagonalDifference` function with the following parameter:

- `int arr[n][m]`: a 2-D array of integers

Return

- `int`: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, `n`, the number of rows and columns in the square matrix `arr`.

Each of the next `n` lines describes a row, `arr[i]`, and consists of `n` space-separated integers `arr[i][j]`.

Constraints

- $-100 \leq arr[i][j] \leq 100$

Sample Input

STDIN	Function
4 5 6 9 8 9	2

```
41 }  
42 else{  
43     diff=sump-sums;  
44 }  
45 return diff;  
46 }
```

Line: 35 Col: 26

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

Test case 0

Test case 1

Test case 2

Compiler Message

Success

